

NOTES ON FLUID PRESSURE

FLUID STATICS-- A resultant problem, resolving the pressure distribution that a fluid exerts on a surface into a single resultant force $\vec{F}_R$. This involves finding the magnitude and location of $\vec{F}_R$. It is a centroid problem.

Defining some units that we will use:

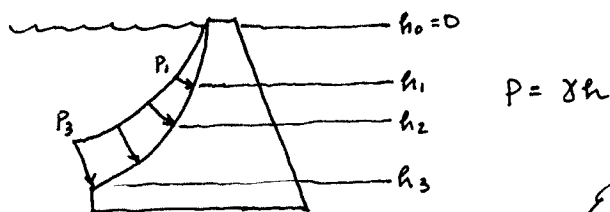
Quantity	Typical Values for Fresh Water	
	SI	US
Mass Density, ρ	$\rho_w = 1000 \frac{\text{kg}}{\text{m}^3}$ $= 1 \frac{\text{Mg}}{\text{m}^3}$	—
Specific Weight, γ	$\gamma = \rho g$ $\gamma_w = 9810 \frac{\text{N}}{\text{m}^3}$	$\gamma_w = 62.4 \frac{\text{lb}}{\text{ft}^3}$
Pressure Gradient	$\frac{9810 \text{ N/m}^2}{\text{m}}$ $= 9.81 \frac{\text{kPa}}{\text{m}}$	$\frac{62.4 \text{ lb/ft}^2}{144 \text{ in}^2/\text{ft}^2}$ $= 0.433 \frac{\text{psi}}{\text{ft}}$

SOME BASIC RULES

① $P = f(\text{depth, Sp. Grav. of fluid})$ $\left. \begin{array}{l} \gamma, \rho \\ = \text{constant} \\ (\text{fluid incompressible}) \end{array} \right\}$

$$P = \gamma h = \rho g h$$

② Pressure at a depth will act $\perp$ to surface.



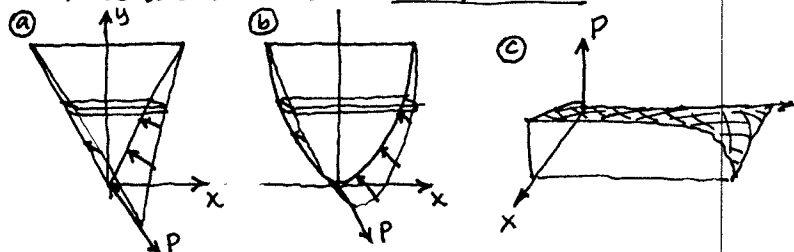
③ Most general eqn to find $|\vec{F}_R|$:

$$|\vec{F}_R| = \int_V dV \quad |\vec{F}_R| = \text{Volume of the pressure distribution}$$

location of $\vec{F}_R$ is $(\bar{x}, \bar{y})$, the centroid of the pressure distribution.

e.g. $\bar{x} = \frac{\int_V \bar{x} dV}{\int_V dV} \quad dV = P dA$

These are used for 3-D problems such as:

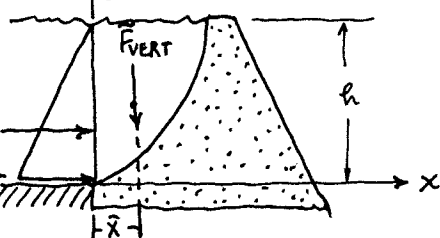


(a) and (b) are the ends of tanks which are non-rectangle.

④ MOST USEFUL EQUATION Used for 2D Problems

Assume a 1m or 1ft wide slice of the pressure distribution. Find $\vec{F}_R$ for that.

$$\vec{F}_R = \vec{F}_{\text{HORIZONTAL}} + \vec{F}_{\text{VERTICAL}}$$

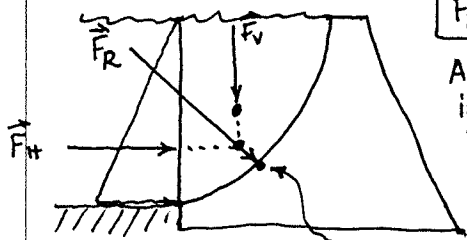


$P_{\text{max}} = \rho g h$
 $= \gamma \cdot h$

$F_H = \frac{1}{2} h P_{\text{max}} \cdot 1$
Knit thickness

$\vec{F}_{\text{HORIZONTAL}} \equiv$ Volume of (horizontally acting) pressure distribution. Acts at centroid ($\frac{1}{3}$ up).

$\vec{F}_{\text{VERTICAL}} \equiv$ Downward force of weight of fluid above structure.
 $= (\text{Volume of fluid}) \cdot \gamma$
 $= (\text{Area})(\text{Unit thickness}) \cdot \gamma$
Acts at centroid of area, $\bar{x}$.



$$\vec{F}_R = \vec{F}_V + \vec{F}_H$$

Acts at intersection of $\vec{F}_H$ and $\vec{F}_V$

$P_1 = \gamma h_1$
 $P_2 = P_{\text{max}} = \gamma h_2$

Magnitude of $\vec{F}_R$:

$$|\vec{F}_R| = \sqrt{F_H^2 + F_V^2}$$

Note: This intersection is NOT the "center of pressure", as defined by your book

Anytime have square plate

$$F_R = \frac{1}{2} (P_1 + P_2) (L) (1) \text{ per unit thickness}$$

location of F_R :

$$\bar{y} = \frac{1}{3} \left[\frac{2(P_1 + P_2)}{P_1 + P_2} \right] (L)$$

Point of contact with wall is the "center of pressure".
 $\vec{F}_R$ is NOT NECESSARILY $\perp$ to wall at this point. In fact, generally it will NOT be.

Fluid Pressure Ex Probs [SI System]

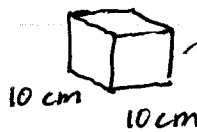
SI Units :
(fresh water)

$$\rho_w = 1000 \frac{\text{kg}}{\text{m}^3}$$

~ Mass Density

$$\rho = 1 \text{ gm/cm}^3$$

$$= 1 \text{ gm/cc}$$



$$10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm} = 10^3 = 1000 \text{ cc}$$

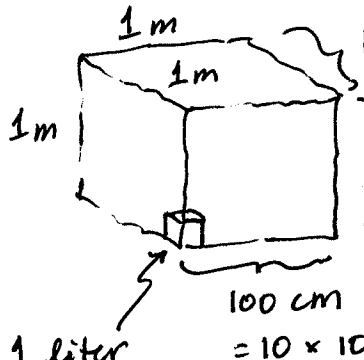
$$= 1 \text{ liter}$$

Has mass = 1000 gm

$$= 1 \text{ kg}$$

$$\rho = 1 \text{ kg/l}$$

"centi" = $\frac{1}{100}$



$$1 \text{ m} = 100 \text{ cm} = 10 \times 10 \text{ cm}$$

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Thus, there are

$$10 \times 10 \times 10 = 1000 \text{ l/m}^3$$

Each liter has

$$\text{mass} = 1 \text{ kg}$$

$$= 10^3 \text{ cm}^3$$

$$= 10 \text{ cm} \times 10 \text{ cm} \times 10 \text{ cm}$$

$$1 \text{ m}^3 \text{ has mass} = 1000 \cdot \text{kg/m}^3$$

$$1 \text{ Mg} = 10^6 \text{ gm} = 10^3 \text{ kg}$$

Thus, fresh water :
(Mass Density, ρ)

$$\rho = \frac{1 \text{ Mg}}{\text{m}^3} = \frac{1000 \text{ kg}}{\text{m}^3}$$

Specific Weight :

$$\gamma = \rho \cdot g = \frac{(9.81)(1000 \text{ kg})}{\text{m}^3}$$

$$\gamma = \frac{9810 \text{ N}}{\text{m}^3}$$

Pascal = P_a

Pressure $P = \rho \cdot g \cdot h = \gamma \cdot h$

e.g. at $h = 10 \text{ m}$, $P = (10 \text{ m}) \left(\frac{9810 \text{ N}}{\text{m}^2} \right) = 98,100 \frac{\text{N}}{\text{m}^2}$

$$= 98.1 \text{ kPa}$$